

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	15 August 2026
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

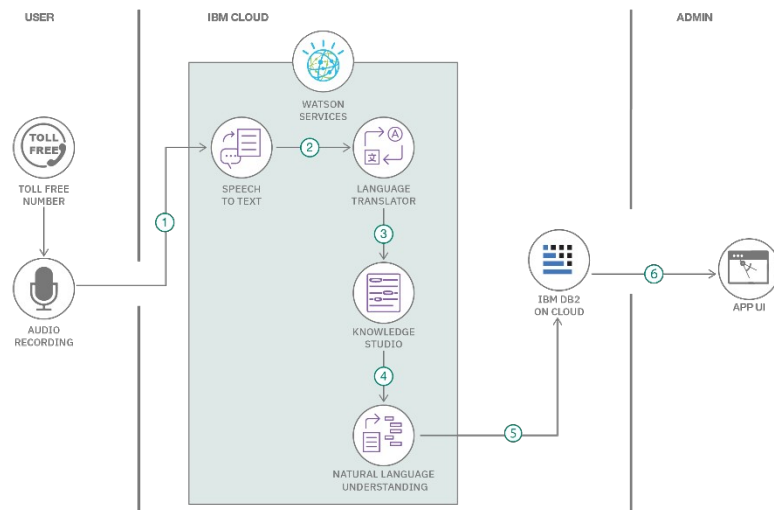
Technical Architecture:

The solution uses ServiceNow as the centralized cloud platform. Users submit a Network Request catalog item through the portal. Service Catalog captures request details, Flow Designer automates approvals and task assignment, network support teams fulfill the generated tasks, and ServiceNow notifications provide status and completion updates.

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Automated Network Request Management in ServiceNow

Reference: ServiceNow Service Catalog and Flow Designer documentation



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Portal interface used by employees to submit network access, connectivity, and service requests.	ServiceNow Service Portal / Employee Center
2	Application Logic-1	Captures request details, validates mandatory information, and creates the catalog request.	ServiceNow Service Catalog, Catalog Item, Variables
3	Application Logic-2	Automates approvals, routing, task creation, and assignment to the appropriate network support group.	ServiceNow Flow Designer
4	Application Logic-3	Manages request status, fulfillment activities, and user notifications.	ServiceNow Notifications, Business Rules, Flow Designer
5	Database	Stores requests, variables, approvals, tasks, users, groups, and configuration records.	ServiceNow Platform Tables / Database
6	Cloud Database	Provides platform-managed cloud storage for application and configuration records.	ServiceNow Cloud Data Storage
7	File Storage	Stores supporting documents and attachments submitted with network requests.	ServiceNow Attachments
8	External API-1	Optional integration with external network or enterprise systems when automated fulfillment requires it.	REST API / ServiceNow IntegrationHub
9	External API-2	Optional interface for external asset, infrastructure, or notification systems.	REST API / IntegrationHub
10	Machine Learning Model	Not required for the core project because request processing is rule- and workflow-based.	Not Applicable
11	Infrastructure (Server / Cloud)	Application is hosted and executed on the ServiceNow cloud platform.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The solution mainly uses ServiceNow configuration and platform capabilities rather than a separate open-source application framework.	ServiceNow Service Catalog, Flow Designer, JavaScript
2	Security Implementations	Role-based access, controlled approvals, mandatory fields, ACLs, and authenticated access protect request and fulfillment data.	ServiceNow ACLs, Roles, Authentication, RBAC
3	Scalable Architecture	The process separates request intake, approval, task assignment, fulfillment, and notification activities, supporting increased request volumes.	ServiceNow Platform, Service Catalog, Flow Designer
4	Availability	The application runs on the ServiceNow cloud platform, providing centralized access and platform-managed availability.	ServiceNow Cloud Platform
5	Performance	Structured catalog forms, automated routing, and workflow-based processing reduce manual effort and improve request turnaround time.	ServiceNow Catalog Item, Flow Designer, Notifications

References:

ServiceNow Service Catalog documentation

ServiceNow Flow Designer documentation

ServiceNow Developer documentation

ServiceNow Platform documentation

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

